

Task:1 Iris Flower Classification using Machine Learning

Git repository link: https://github.com/guduguntlakeerthana/CodeAlpha_Iris-Flower-Classification

Aim

To classify Iris flowers into three species:

- Iris Setosa
- Iris Versicolor
- Iris Virginica

using machine learning algorithms based on flower measurements.

Introduction

Iris Flower Classification is a basic machine learning classification project.

The dataset contains measurements of iris flowers such as:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Using these measurements, a machine learning model predicts the species of the flower.

This project helps in understanding:

- Supervised Learning
 - Classification
 - Model Training
 - Model Testing
 - Accuracy Evaluation
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Tools and Libraries Used

- Python
- Pandas

- NumPy
- Matplotlib
- Scikit-learn

Official library website:

[Scikit-learn](https://scikit-learn.org/)

Dataset Description

The Iris dataset contains 150 flower samples.

There are 3 classes:

1. Iris Setosa
2. Iris Versicolor
3. Iris Virginica

Each flower sample has:

- Sepal Length
 - Sepal Width
 - Petal Length
 - Petal Width
 - Species
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Machine Learning Steps

1. Import libraries
 2. Load dataset
 3. Split dataset into training and testing data
 4. Train the model
 5. Predict results
 6. Evaluate accuracy
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Python Program

```
# Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report

# Load dataset
data = pd.read_csv("Iris.csv")

# Display first 5 rows
print(data.head())

# Selecting input features
X = data[['SepalLengthCm', 'SepalWidthCm',
          'PetalLengthCm', 'PetalWidthCm']]

# Selecting target output
y = data['Species']

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Create model
model = KNeighborsClassifier(n_neighbors=3)

# Train model
model.fit(X_train, y_train)

# Predict output
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

# Print accuracy
print("Accuracy:", accuracy)

# Classification report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

Output

Example Output:

Accuracy: 1.0

Classification report:

	precision	recall	f1-score
Iris-setosa	1.00	1.00	1.00
Iris-versicolor	1.00	1.00	1.00
Iris-virginica	1.00	1.00	1.00

Explanation of the Code

1. Import Libraries

Libraries required for data handling and machine learning are imported.

2. Load Dataset

The Iris.csv dataset is loaded using Pandas.

3. Feature Selection

Input features:

- SepalLengthCm
- SepalWidthCm
- PetalLengthCm
- PetalWidthCm

Target:

- Species

4. Train-Test Split

- 80% data used for training
- 20% data used for testing

5. Model Creation

K-Nearest Neighbors (KNN) classifier is used.

6. Model Training

The model learns patterns from training data.

7. Prediction

The model predicts flower species from test data.

8. Accuracy Evaluation

Accuracy score checks how correctly the model predicts species.

Advantages

- Simple and easy to understand
 - Good project for beginners
 - Demonstrates supervised learning clearly
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Applications

- Pattern recognition
 - Botanical research
 - Species identification
 - Machine learning education
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Conclusion

The Iris Flower Classification project successfully classifies iris flower species using machine learning techniques. The KNN model provides high accuracy and helps in understanding basic classification concepts such as training, prediction, and evaluation in machine learning.